

Ansh Bhansali

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EDUCATION

University of Illinois Urbana-Champaign, IL, USA Aug 2025 – Dec 2026
Master of Engineering in Autonomy and Robotics | GPA: 3.9/4.0

Sardar Patel Institute of Technology, Mumbai, India Dec 2021 – Jul 2025
Bachelor of Technology, Electronics and Telecommunication | GPA: 3.9/4.0

EXPERIENCE

Robotics Engineering Intern - AgRobotics, CA, USA Aug 2026 – Present
Project: Manipulation and Navigation for Agriculture Robots

Research Engineer - IHSI, University of Illinois, IL, USA Apr 2026 – Aug 2026
Project: ARPA-H MarginDx, AI-Guided Robotic Microsurgery for Tumor Margin Assessment

- Developed real-time **impedance force control** stack for **KUKA LBR Med 7 + PI hexapod** achieving tissue contact regulation.
- Built multi-layer IPC architecture (**Java/Python/UDP**) bridging KUKA, PC, and hexapod, with **sub-10ms** command latency.
- Achieved **10 μ m** resolution 3D tissue surface reconstruction with stable contact force during raster scans on tissue phantoms.

Embedded Engineering Intern - Dimension Six Technologies, Mumbai, IN May 2024 – May 2025
Project: STM32-based E-Bike Conversion Kit

- Engineered production-ready firmware with battery management and regenerative braking, extending vehicle range by **40%**.
- Designed a 4-layer PCB in KiCad for a custom **ESC**, optimizing power stage layout to reduce losses under peak load.
- Deployed IoT stack on **ESP32S3** with RFID authentication, enabling remote monitoring and contactless payments over MQTT.

Robotics Research Intern - Indian Institute of Technology, Bombay, Mumbai, IN Jan 2024 – Jun 2024
Project: Autonomous Rugged Robot for Military Applications

- Built **SLAM**-based autonomous navigation stack with path planning, achieving **95%** accuracy in dynamic indoor environments.
- Fused IMU, GPS, and RGB-D camera data via **Extended Kalman Filter**, improving localization accuracy.
- Deployed **YOLOv3** for real-time human detection and designed an adaptive gait controller for robust stair-climbing traversal.

PROJECTS

O-VLA Trajectory Optimization Safety Layer (*ProxQP, CHOMP, RNEA, Pinocchio*) ([Github](#))

- Built **morphology-agnostic** trajectory optimizer transforming unsafe VLA outputs into collision-free, constraint-satisfying paths.
- Achieved **100%** constraint satisfaction and collision reduction across 5 robot morphologies in **0.03-0.44s**.
- Demonstrated **34-45%** energy reduction through coupled dynamics optimization using **Recursive Newton-Euler Algorithm**.

Humanoid Whole-Body Motion Planning (*RL, MPC, Unitree G1, MuJoCo*) ([Github](#))

- Implemented motion planning stack using **ZMP** preview control, **A*** footstep planning, and **MPC** balance optimization
- Achieved **49%** lower energy use than PD control with MPC balance control, while remaining stable under **70-80N** perturbations.
- Integrated **RL**-based locomotion with Jacobian IK manipulation for coordinated walk-and-reach tasks.

VLM-Based Autonomous Navigation System (*GEM e4, GPT-4o, ROS2*) ([Github](#))

- Built autonomous navigation stack integrating **GPT-4o VLM** for intent-based planning with LiDAR-camera sensor fusion.
- Implemented full pipeline: BEV perception, A* path planning, and Stanley control achieving end-to-end task success.
- Achieved **88%** accuracy on KITTI real-world dataset for VLM-based goal localization from natural language commands.

Open-Vocabulary 6D Pose Estimation (*FoundationPose, SAM-3, Moondream2*) ([Report](#))

- Developed **zero-shot** 6D tracking architecture with VLM-based **semantic inventory** and Gemini API query enrichment.
- Integrated **SAM-3** prompt segmentation with **Objaverse** mesh retrieval, achieving **76-100%** ADD-S AUC on YCBV dataset.
- Built RGB-D variant using **LangSAM** with **InstantMesh** and median-volume ensemble selection for mesh consistency.

SKILLS

- Languages:** Python, C++, MATLAB, Java, Bash
- AI & Perception:** VLA Models, Transformers, PyTorch, RL, Behavior Cloning, Diffusion Policy, Computer Vision, Pose Estimation
- Planning & Control:** Motion Planning, Trajectory Optimization, Whole-Body Control, Sim-to-Real Transfer, SLAM, Sensor Fusion
- Tools & Frameworks:** ROS2, MuJoCo, Isaac Lab, Pinocchio, MoveIt, Behavior Trees, Task Planning, Git, Linux